

**BEFORE THE NATIONAL GREEN TRIBUNAL
PRINCIPAL BENCH, NEW DELHI**

Original Application No.985/2019

In Re: Water Pollution by Tanneries at Jajmau, Kanpur, Uttar Pradesh

WITH

Original Application No.986/2019

In Re: Water Pollution at Rania, Kanpur Dehat & Rakhi Mandi, Kanpur
Nagar, Uttar Pradesh

With

Original Application No.528/2023

In re: news report published in Dainik Jagran dated 14.08.2023
**“highlights a growing concern regarding industrial pollution
in the Godhrauli village”**

Date of hearing: 29.11.2023

**CORAM: HON’BLE MR. JUSTICE PRAKASH SHRIVASTAVA, CHAIRPERSON
HON’BLE MR. JUSTICE SUDHIR AGARWAL, JUDICIAL MEMBER
HON’BLE MR. JUSTICE ARUN KUMAR TYAGI, JUDICIAL MEMBER
HON’BLE DR. A. SENTHIL VEL, EXPERT MEMBER**

Applicant(s): Ms. Katyayni, Adv. (Amicus Curiae) in OA 985-986/2019
Respondent: Mr. Pradeep Misra & Mr. Daleep Dhyani, Advs. for UPPCB in OA 985-
986/2019 (Through VC)
Mr. Raj Kumar, Adv. for CPCB in OA 985-986/2019 (Through VC)
Ms. Priyanka Swami, Adv. for the State of UP in OA 985-986/2019

ORDER

**Original Application No. 985/2019, Original Application No.
986/2019 and Original Application No. 528/2023**

1. The issue involved in these OAs are firstly failure to scientifically handle and dispose of Chromium dumps at Rania, Kanpur Dehat and Rakhi Mandi, Kanpur Nagar which are in existence since 1976 and have caused contamination of ground water (O.A. 986/2019) and secondly, continuing water pollution by tanneries discharging untreated/partially

treated industrial effluents, containing toxic Chromium into the irrigation canal through inadequately functioning CETP at Jajmau, UP (i.e. OA 985/2019).

2. In O.A. 528/2023- the issue involved is in respect of industrial pollution and discharge of chromium into the ground water by industrial units at Godhrauli Village near Kanpur adversely affecting eight neighboring villages namely, Aung, Baniyan Khera, Rawatpur, Rampur, Pahur, Shukla Nagar, Akbar Khera, Kchakpur-Gadi.

3. The Tribunal in the order dated 16.05.2023 passed in OA No.985/2019 and OA No.986/2019 had taken note of its earlier orders and had noted compliance status after the previous orders upto that date as under:-

“8. In pursuance of above, reports have been filed by State PCB (21.02.2023) and Oversight Committee (22.02.2023, 24.03.2023 and 09.05.2023).

9. The report of the State PCB is that MoUs have been executed between the TSDF operators - M/s U.P. Waste Management Project & M/s Bharat Oil and Waste Management Ltd., Kumbhi, Kanpur Dehat and Additional District Magistrate(Admin) Kanpur Dehat on 05.11.2022 and work orders have been issued to the TSDF operators by Additional District Magistrate (Admin) Kanpur Dehat on 15.11.2022 for lifting, treatment and scientific disposal of chromium waste as per the rate decided by the committee constituted by Environment, Forest and Climate Change Department Government of U.P. Both the TSDF operators were directed to start the work from 17.11.2022. Both the TSDF operators started the work regarding completing the necessary formalities before start of the work such as construction of tyre washing platform, dust suppression arrangements, arrangement of weigh bridge, route chart for transportation of Chromium waste from dump site to TSDF facility, Demarcation of site for lifting the waste, fencing of the site by green net, installation of PTZ camera with connection to UPPCB's monitoring portal and construction of site office etc. It has been further stated:-

“5. That it is submitted that a meeting of Committee and Sub Committee was again held on 22.11.2022 in hybrid mode to review the progress regarding completion of necessary formalities and both the TSDF operators were directed to complete all the formalities as per MoU at the earliest. Further a meeting with all the stakeholders was held under chairmanship of ADM (Admin), Kanpur Dehat on 06.12.2022 to further review the progress, in which officers of weight and

Measure Department, Toll Plaza officers of Kanpur-Itawa Highway, Public Work Department, Regional Office, U.P. Pollution Control Board and both the operators of TSDFs. In the meeting both the operators have informed that they have completed all the formalities.

6. That it is submitted that lifting of Chromium waste from the site to TSDFs started on 15.12.2022 and it is going on. Committee has ensured the scientific disposal of waste and suitable SoP has been devised by for the same. At the TSDFs, the weight of chromium waste are again measured and stored separately before stabilization. **After stabilization the sample of the stabilized waste is collected and sent to Laboratory of Central Pollution Control Board, Delhi for TCLP test and after the results are found as per the prescribed standards for disposal in SLF, the operators are allowed to dispose in the SLF. The quantity of Chromium waste lifted from the site is also monitored through a mobile app developed by U.P. Pollution Control Board. Both the TSDF operators have lifted approx. 28596 MT chromium waste from dump site till 20.02.2023.**
7. That Hon'ble NGT has directed UPPCB and CPCB to find out suitability of setting up of pilot plant model to treat chromium by way of setting up of RO/Ultra/Nano Filtration process and application of Biotechnology to remove chromium as being done at other places for removal of arsenic/fluoride. That it is submitted that in compliance of above direction a meeting was held through hybrid mode on 11.01.2023 regarding remediation of chromium contaminated ground water at Khanchandpur Rania, Kanpur Dehat. During meeting it was also informed that ground water remediation technology suggested in earlier approved DPR under NCEF project was pumped and Ex-situ treatment. **Further strategy for remediation can be decided after present condition of ground water quality and extent/level of contamination at Chromium contaminated sites at Rania, Kanpur Dehat after monitoring of ground water from old /existing monitoring wells.** Minutes of the first meeting held on 11.01.2023 and communicated vide letter dated 06.02.2023 are being attached herewith and marked as **Annexure No.-3**
8. That it is submitted that in the State of Uttar Pradesh, **similar nature of ground water contamination containing hexavalent chromium is being remediated at Lohia Nagar, Ghaziabad through pump and treat method along with combination of treatment through bio-culture/micro organisms. So, technology already exists.** A brief status regarding the technology is attached herewith and marked as **Annexure No. -4**
9. That it is submitted that U.P. Pollution Control Board had imposed Environmental Compensation of Rs. 280.01 Crore against 08 defaulting industries on 14.04.2022. Hon'ble NGT

has revised the amount of Environmental compensation imposed by UPPCB, considering the appeal filed by industries vide order dated 05.12.2022, 15.12.2022, & 16.01.2023 in Appeal Nos. 14, 15, 16, 17 and 18 of 2020. The detail of revised Environmental Compensation is as follows :

Sr.No.	Name & Address of industry	Environmental compensation amount imposed by the State Board (Rs.)	Revised environmental compensation amount by Hon'ble NGT (Rs.)
1	Chandani Chemicals Pvt. Ltd. Khanchandpur , Rania, Kanpur Dehat	27,40,62,295.00	23,09,31,000.00
2	HeilgerChem Pvt. Ltd. VIII.- Chiraura, Raipur, Rania, Kanpur Dehat	39,62,13,484.00	25,52,34,375.00
3	Rukmani Chemicals Pvt. Ltd. Rania, Kanpur Dehat	46,89,51,039.00	13,69,56,375.00
4	Amelia Textiles & Chemicals Pvt. Ltd. 23, Khanchandpur, Rania, Kanpur Dehat	14,61,66,558.00	11,92,50,000.00
5	Waris Chemicals Pvt Ltd. Khanchandpur, Rania, Kanpur Dehat	44,34,53,042.00	25,39,68,750.00
6	Cerulean Chemicals Pvt. Ltd. Khanchandpur, Rania, Kanpur Dehat	87,69,99,345.00	-
7	Kaleena Chemicals, Pvt Village Malo, GT Road, Chaubepur, Kanpur Nagar	18,02,72,088.00	-
8	Bharat Chemical Udhyog C/O Smt Tarannum Fatima W/o Shri Sahabuddin and Shri Sahabuddin S/o Mohd Nafees, 39/72, Majeed Ahmad Road, Kanpur Nagar and ShriIftekharul Amin, ShriIqbal Ahsan, ShriWaqarul Amin	1,40,07,150.00	-

10. That it is submitted that U.P. Pollution Control Board has also given direction vide letter dated 07.02.2023 to the concerned industries to deposit the environmental compensation as per the orders of Hon'ble NGT dated 05.12.2022, 15.12.2022, & 16.01.2023 in Appeal Nos. 14, 15, 16, 17 and 18 of 2020 with copy to District Magistrate, Kanpur Nagar for necessary action. The environmental compensation has not been deposited by any industry till now. The copies of letter dated 07.02.2023 are enclosed herewith and marked as **Annexure No.-5**

11. That it is submitted that **U.P. Pollution Control Board is in the process to invite EoI also from various stakeholders to submit draft pilot plant project report for deciding possible technologies for ground water**

remediation at Khanchandpur Rania, Kanpur Dehat. That it is submitted that the work of ground water remediation could only be started after removal of lying chromium waste and stabilization of contaminated soil, so that their effect is ruled out from ground water. Latest ground water analysis is also required to be carried out to ascertain the extent of contamination of ground water in that area.

12. *There are 400 tanneries in Jajmau Kanpur which are established in approximately 7.431 sq. kilometre area within 50 meter to 1.0 Km distance from river Ganga. The smallest tannery is of 100 sq. meter area and largest is of 34500 sq. meter area. The tanneries are established since 1946 at Jajmau. Most of the tanneries are small scale tanneries and have only installed primary effluent treatment plant and the treated effluent is being treated through a common effluent treatment plant installed by State Government through indodutch assistance in 199293. That it is further submitted that there is no separate sewerage network in Jajmau area, where all the tannery units are situated, so **untreated domestic sewage generated from Jajmau area also gets mixed with industrial effluent which goes through conveyance channel into CETP.***
13. *That it is submitted that the **CETP at Jajmau, Kanpur is having capacity of 36 MLD to treat (combined effluent in ratio of 1:3) 9 MLD Tannery effluent & 27 MLD domestic sewage effluent. All the tannery units discharge primary treated effluent through conveyance channel, which is further pumped through 4 pumping stations into 36 MLD CETP designed to treat mixed effluent consisting of 9 MLD tannery effluent and 27 MLD domestic sewer.***
14. *That the maintenance of conveyance channel, pumping stations and operation of CETP is done by Ganga Pollution Control Unit, U.P. Jal Nigam, Kanpur. The **treated CETP effluent gets further diluted with treated effluent of 130 MLD, 43 MLD and 5 MLD STP at the premises of CETP Jajmau before finally discharged into irrigation channel for irrigation. The STPs are also operated and maintained by operator M/s Kanpur River Management Pvt Ltd, Kanpur selected by UP Jal Nigam.***
15. *That, it is submitted that **in order to control industrial pollution from tannery sector in Jajmau area, State Government has devised a short term and a long term plan so that pollution of river Ganga due to effluent from tanneries is completely stopped.***
- A. *That as a **Short Term Plan**, the strategy adopted is the improvement in compliance through enforcement measures as detailed below:*

- a. *That U.P. Pollution Control Board issued guidelines vide letter dated 08.06.2019 to assess the infrastructure of tannery units according to their permitted capacity in order to restrict tannery units to reduce the chances of overflow from pumping stations as well as excess effluent going into CETP.*
- b. *That it is submitted that UPPCB has sealed 169 drums & 47 Paddles in 92 tannery units associated with Pumping Station-03 (Wajidpur), 54 drums & 12 Paddles in 19 tannery units associated with Pumping Station-04 (Budhiyaghat), 13 drums & 07 Paddles in 12 tannery units associated with Pumping Station-01 (Chhabilepurwa) to control excess flow generation.*
- c. *That U.P. Pollution Control Board restricted the overall processing capacity of each tannery unit further to 50% of its consented capacity in order to reduce the chances of excess flow going to pumping stations and subsequently to CETP inlet.*
- d. *That U.P. Pollution Control Board implemented roster of operation of tannery units attached to different pumping stations. As per this roaster, tannery units attached to pumping station 1 and 3 are operated for first 15 days of the month and tannery units attached to pumping station 2 and 4 are operated for next 15/16 days of the month.*
- e. *That U.P. Pollution Control Board has informed that Surprise checks are conducted by UPPCB and Inter Departmental Teams as well as Third Party Institutions like IIT, HBTU, PCCRI, NEERI from time to time and actions are taken against the defaulter units as well as operator of CETP. The details of action taken in form of closure of tannery units is enclosed herewith and marked as **Annexure No. - 6.***
- f. *That U.P. Pollution Control Board has imposed the environmental compensation against the defaulting units from time to time. Environmental Compensation of Rs 2.81 Crore imposed against 259 defaulter tannery units, out of which Environmental Compensation of Rs. 1.22 Crore has been deposited by 240 tannery units of Jajmau Kanpur. Recovery Certificate for recovery of Environmental Compensation has been issued by District Administration against 19 units.*
- g. *That U.P. Pollution Control Board has prosecuted 65 defaulter tannery units of Jajmau Kanpur under section 41/44 of Water (Prevention and Control of Pollution) Act 1974 in Special Court (Air and Water Pollution Control) Lucknow. Out of which 36 cases have been decided and conviction is done in 26 cases.*
- h. *That U.P. Pollution Control Board has imposed Environmental Compensation against the operator of*

CETP and STPs at Jajmau Kanpur i.e. Ganga Pollution Control Unit, U.P. Jal Nigam, Kanpur / M/s Kanpur River Management Pvt Ltd, Kanpur from time to time on the basis of different defaults. The detail of imposition of Environmental Compensation is enclosed herewith and marked as **Annexure No. -7.**

- B.** That as a **Long Term Plan**, the strategy adopted involves construction of a new CETP and upgradation of existing PETPs along with adoption of cleaner technology.

That it is submitted that in order to improve the treatment capacity of existing tannery units of Jajmau, Kanpur, Central Government has taken effective initiation and for Jajmau Tannery cluster, **National Mission for Clean Ganga, New Delhi has approved 20 MLD CETP (dilution based) worth Rs. 554 Crore, as per Hon'ble NGT direction. The proposal was consented by the tannery association and the Government of Uttar Pradesh. The member tannery units have committed to contribute Rs. 25 crore and has proposed to bear the estimated cost of Rs. 137 Crore for upgradation of their Primary Effluent Treatment Plants (PETPs). Rest 392 Crore will be borne by National Mission for Clean Ganga (NMCG). The Special Purpose Vehicle "Jajmau Tannery Effluent Treatment Association (JTETA)" of member tannery units and district administration headed by District Magistrate, Kanpur Nagar has been formed.**

That the Jajmau Tannery Effluent Treatment Association (JTETA) has allotted the work of construction of CETP to M/s VA Tech Wa Bag Ltd. after finalizing the tender with approval from NMCG. NMCG has also appointed construction engineer to supervise the ongoing work and estimated time of completion is 13.08.2023. The new CETP is being constructed in the same premises of existing CETP and the treated effluent is proposed to be used for irrigation after dilution with treated STP effluent. NMCG has also reviewed the progress of installation of 20 MLD CETP on 06.02.2023 with concerning stakeholders.

After the construction of new 20 MLD CETP & allied structures like closed conveyance system to carry primarily treated effluent from member tannery units to CETP & CCRP and upgradation of existing PETP by tanneries, complete elimination of accidental discharge in river Ganga would be ensured.

16. That it is submitted that Central Pollution Control Board has prepared a "Charter for Water Recycling and Pollution Prevention in Tanneries" in order to reduce the TDS in effluent generated from the tannery units by adopting cleaner technology and to minimize the quantity of waste. **Central Pollution Control Board has given various directions to U.P. Pollution Control Board vide letter dated**

18/23.08.2022. In compliance of above, UPPCB has issued directions vide letter dated 02.09.2022 under section 33-A of Water Act 1974 to all tannery units, for adopting cleaner technology to minimize effluent quantity and use of chemicals as per the "Charter for Water Recycling and Pollution Prevention in Tanneries" in time bound manner and submit action plan for the same. Central Leather Research Institute (CLRI) has given license to 43 tannery units of Jajmau for use of Technology regarding waterless chrome tanning process. In continuation of adopting cleaner technology, M/s Solidaridad Asia (having technical collaboration with Neatherlands Enterprise Agency) has installed brush type desalting machine in 18 tannery units, fitted water meter in 30 drums of 20 tannery units, installed drum screen in 01 tannery unit, Screen Bar in 25 tannery units and tallow plant in 01 tannery unit of Jajmau, Kanpur. Besides above 2500 square feet tiles have been constructed from lime sludge obtained from 05 tannery units of Jajmau, Kanpur. Upgradation of PETP in 10 tannery units is also proposed by Solidaridad Kanpur Centre in Jajmau, Kanpur. The member tannery units have also submitted undertaking for upgradation of PETP before commissioning of the new 20 MLD CETP.

That it is submitted that a workshop at Kanpur Nagar has also been organized by CPCB and UPPCB with stakeholders and cleaner technology service providers regarding adoption of cleaner technology on 07-08 February 2023."

10. Report of the Oversight Committee dated 22.02.2023 is as follows:-

Sl No	Directions issued by the Hon'ble NGT	Action taken
1.	Contaminated sites need to be remediated in terms of removing of the waste and transporting to TSDF, followed by due stabilization/ immobilization.	Scientific disposal of the chromium dump has started on 15.12.2022 in Khanchandpur Rania, Kanpur Dehat through two TSDFs, namely, (i) M/s Uttar Pradesh Waste Management Project, a Division of Re Sustainability Ltd, formally known as Ramky Enviro Engineers Ltd, & (ii) M/s Bharat Oil and Waste Management Ltd, Kumbhi, Akbarpur, Kanpur Dehat having processing capacities of 685 Mt/d and 100 Mt/d respectively. Quantities remediated till 31st January 2023 through these two TSDFs are (i) 12126.27 Mt and (ii) 1496.97 Mt respectively (total 13623.24 Mt).The entire quantity of waste is scheduled to be disposed of by May, 2023. The work-plans submitted by the TSDFs are being annexed herewith as Annexure-1.
2.	Contaminated chrome bearing ground water needs to be remediated by appropriate methodology	The UPPCB has informed that it is in the process of inviting EoI from various stakeholders for the submission of draft pilot

	and plan executed in a time bound manner. UPPCB and CPCB may immediately find out suitability of setting up of pilot plant model to treat chromium by way of setting up of RO/ Ultra/ Nano-Filtration process and application of Biotechnology to remove chromium as being done at other places for removal of arsenic/fluoride. This may be done in next three months and the cost of such demonstration/ model project may be borne through EC funds of the UPPCB.	plant project report with a view to deciding the possible technological remedies for ground water remediation at Khanchandpur, Rania, Kanpur Dehat. The work of ground water remediation could be started only after removal of lying chromium waste and stabilization of contaminated soil.
3.	CPCB needs to get its directions dated: 18/23.08.2022 implemented on priority apart from action referred to in Para 11(1) above and also on other issues in a time bound manner.	The UPPCB has informed that: (a) it has issued a letter dated 02.09.2022 with directions under Section 33-A of the Water Act 1974 to all tannery units for adopting cleaner technology with a view to minimize effluent quantity and use of chemicals as per the “Charter for Water Recycling and Pollution Prevention in Tanneries” in a time bound manner. Subsequently, directions for closure and disconnection of electricity of 213 tannery units were issued on 15.09.2022. However, no information on actual closure of these units and disconnection of electricity has been furnished. (b) Consequent upon issue of the letter dated 02.09.2022, Environmental Compensation has been imposed on the following defaulters on different dates: (i) Rs. 48.375 lacs on GPCU, UP Jal Nigam (Urban) for the CETP Jajmau, Kanpur Nagar; (ii) Rs. 21.0 lacs on the KR MPL Agency for the STP Jajmau, Kanpur Nagar; and (iii) Again, Rs. 14.20 lacs on the KR MPL Agency for the STP Jajmau. However, information on recovery of ECs imposed in the above and previous cases has not been shared. (c) The Central Leather Research Institute (CLRI) has granted license to 43 tannery units at Jajmau for use of Technology for waterless chrome tanning process. M/s Solidaridad Asia (in collaboration with Neatherlands Enterprise Agency) has installed brush type desalting machine in 18 tannery units, fitted water meter in 30 drums of 20 tannery units, has installed drum screen in 01 tannery unit, screen bar in 25 tannery units and tallow plant in 01 tannery unit at Jajmau, Kanpur. In addition, 2500 square feet tiles have been constructed from lime sludge obtained from 05 tannery units of Jajmau, Kanpur. Upgradation

		of PETP in 10 tannery units is proposed by M/s Solidaridad, Kanpur Centre in Jajmau, Kanpur. The member tannery units have also submitted undertaking of upgradation of PETP before commissioning of the new 20 MLD CETP.
4.	UPPCB needs to monitor remedial measures to prevent pollution from tanneries in realistic manner issuing guidelines, roaster operations, sealing of drums/pedals, cutting down the production capacities, closing the tanneries till compliance is achieved.	The UPPCB has informed that in order to control industrial pollution from tannery sector, it has issued several instructions on roaster operations, sealing of drums/pedals, cutting down the production capacities and, closure orders against the non-compliant tanneries. It also undertakes enforcement measures regularly. However, as per the data available on the UPECP web portal regarding water quality of river Ganga at Kanpur, its upstream and downstream continues to remain in D Category. Further, the water quality of river Ganga at downstream of Kanpur is worse than the upstream indicating that these units and other wastes add to the pollution of the river Ganga in this stretch. For details the Annexure-2 may kindly be referred.
5.	NMCG needs to ensure that CETP at Jajmau is completed as already long delay has taken place.	As per information received from the NMCG 66.95% construction work of the 20 MLD CETP at Jajmau Kanpur, has been completed till 31st January 2023. The targeted time for completion has been reported as August 2023. Having regard to the pace of work in past about 18 months, it does not appear to be realistic. Till June, 2021, the progress was about 30%. In February, 2022 it became 43.5%, in June, 2022 it was 54% and in January, 2023, it is reported as 67%. On each of these reporting dates, except the last one, an unrealistically closer target date was given. This has been pointed out in earlier reports of this Committee. In last 07 months, only 13% progress has been made. At the present pace, it appears difficult to complete the project by August, 2023.

11. Report of the Oversight Committee dated 09.05.2023 mention the compliance states as follows:-

“Status of Compliance:

Progress regarding various directions issued by the Hon’ble NGT is same as reported in the previous reports except on direction no. 01 where it has been reported that the M/s Uttar Pradesh Waste Management Project has lifted 72,283 MT and treated 57,744.645 MT out of 80,000 MT of allocation of the waste and M/s Bharat Oil and Waste Management Ltd has lifted 4962 MT and treated 2898.54 MT out of 5000 MT of allocation of waste. Considering the pace of work and the quantity of waste remediated till date, scientific remediation and disposal of the entire waste, appears to be achievable by the target date.”

4. In the order dated 16.05.2023, the Tribunal had reached to the following conclusions:-

- “13. We have considered the matter further with the help of learned Amicus, Ms. Katyayni and learned Counsel for the State PCB. The comments of the Minister of State for Women Welfare and Child Development, UP, need verification by the State PCB and Oversight Committee with reference to ground situation and remedial measures taken.
 14. We find that progress in construction work of CETP at Jajmau is not satisfactory. Further, water quality of river Ganga and Kanpur Nagar which still remains in D-category needs to be improved by preventing discharge of pollution upstream. CETP needs to remediate Chromium, Chromium need to be treated in Chromium Recovery Plant so to ensure that, chrome bearing effluents are not discharged into river Ganga or loaded on land on the name of irrigation by exploiting dilution by treated sewage effluents. The next report needs to cover the issue of existing chromium level in ground water where currently such combined effluents are used for irrigation. After installation of 20 MLD CETP, a well laid down monitorable ferti-irrigation plan will have to be laid down notifying the periods in which such effluents will be let out in Ganga and their quality. This aspect be overseen by State PCB, CPCB and NMCG.
 15. Periodical health surveys be undertaken and remedial action taken in terms of safe drinking water in light thereof.
 16. Further, we note that out of estimated 85,008 MT of legacy chromium waste at different locations, two companies have lifted 77,245 MT of waste and taken to the TSDF site for treatment and stabilization as per the Agreement and the work order referred to above. Work of transportation and stabilization was to be completed by 16.05.2023 i.e. today. In order to ensure that due SOPs are followed during excavation, transportation, and stabilization, UP PCB needs to monitor the process as per authorization granted.
 17. With regard to remediation of ground water contamination, we do not find current status of ground water quality profile and nor of pilot project to show removal of chromium by installing RO and nano filtration system, as directed vide order dated 23.09.2022. While pump and treat method along with bio-remediation may be long term approaches, measures like RO installation at small level can quickly demonstrate performance which may be done on priority. Possibility of stabilised waste being converted into concrete blocks may be explored.”
5. In the aforesaid background, Tribunal on 16.05.2023 had issued the following directions:-
18. In the light of above, further status report in the matter be filed with regard to status as on 31.10.2023 by joint Committee of NMCG, State PCB, CPCB and District Magistrate, Kanpur Dehat on or before 15.11.2023 by e-mail at judicial-ngt@gov.in preferably in the form of searchable PDF/ OCR Support PDF and not in the form of Image PDF. State PCB will be the nodal agency for coordination and compliance. The report may inter alia mention:-

- i. *Status of complete removal and transportation of excavated chromium waste from all the sites and waste treated and stabilized and duly ensured for stability before landfilling.*
- ii. *ground water profile in terms of chromium in the areas under question and installation of pilot project on chromium removal by RO/Nanofiltration.*
- iii. *Execution plan with timelines on ground water remediation.*
- iv. *Completion of CETP with respect to chromium treatment and well laid ferti-irrigation plan, existing ground water scenario and periods of discharging effluents into Ganga.”*

6. In OA No. 528/2023, the Tribunal in the previous order dated 21.08.2023 had taken note of the fact that the factories are recklessly discharging untreated effluents into the ground through boreholes and that water from a few handpumps had a yellow hue. The Tribunal had enumerated the steps which are required to be taken by the statutory authorizes as under:-

“6. In view of the above, following steps are required to be taken by the statutory authorities:

- i. *Assess the ground water quality of the villages in question.*
- ii. *Ascertain the cause of ground water contamination and the reasons including ignoring the cause.*
- iii. *Actions taken against defaulting agencies – industries and other.*
- iv. *Reporting on ill health effects on villagers/agriculture, if any.*
- v. *Medical facilities/test by medical camp in the villages.*
- vi. *Availability of drinking water in the affected areas, persons, family/villagers.*
- vii. *Inspection of the industrial unit and the investigation of reverse boring as reported and taking further necessary punitive legal actions.*
- viii. *Remedial action taken and to be taken.”*

7. The Tribunal accordingly had formed the joint committee and issued following directions to the joint committee which are as under:-

“9. The Committee is further directed to

- (i) *examine the ground water profile in terms of chromium in the areas under question and taking action for installation of pilot*

project on chromium removal by nano-filtration or other available methods.

- (ii) *Execution of plan with timelines on ground water remediation.*
- (iii) *Taking action as per para 6 above including treatment of chromium and submit report within two months by e-mail at judicial-ngt@gov.in preferably in the form of searchable PDF/OCR Support PDF and not in the form of Image PDF.”*

8. In terms of the directions of the Tribunal in OA No.985/2019 and OA. No. 986/2019 compliance report by UPPCB dated 24.11.2023 has been filed. The report states that UPPCB has constituted a committee for verification of the facts mentioned in the letter dated 27.09.2022 of Ms. Pratiba Shukla. Further, states that lifting of chromium waste has started on 15.12.2022 and that ADM (Admin), Kanpur Dehat had issued the work order to TSDF operators for lifting, treatment and scientific disposal of 85006 MT chromium waste on 15.11.2022 and the same has been lifted by TSDF operators from the dump site to TSDF. The report discloses that on the inspection by officials of CPCB and UPPCB of the dump site on 19.05.2023 approximately 7000-10000 MT of hexavalent chromium waste was found at the site. For this, the work order was issued to TSDF operators. Ground water samples of Khanchandpur, Rania, Kanpur Dehat taken in pre monsoon period, the presence of Cr(VI) was found and for post monsoon samples, the report is stated to be awaited. Paragraph 11 of the report states that proposal for stabilization of soil at the site received from Maa Institute of Environmental Biotechnology Research and Services (MIEBRS), New Delhi, is under consideration. The Counsel for the UPPCB has submitted that entrusting the work to a government agency instead of the private agency will be duly considered.

9. The report states about the work of ground water remediation and organizing health camps. As per the report, though, environmental

compensation has been imposed upon the defaulting industries but the amount has not been recovered till now.

10. The report further states that there are 400 tanneries in Jajmau Kanpur established in approximately 7.431 sq. kilometer area within 50 meter to 1.0 km distance from river Ganga, hence, UPPCB may disclose in the next report if these units are operating in flood plain zone of river Ganga.

11. As per the report there is no separate conveyance system for effluents from tanneries and sewage network in Jajmau area, therefore, untreated domestic sewage get mixed with the industrial effluent which goes through the conveyance channel to the CETP, STP and through untapped drains to river Ganga. For Jajmau following short-term and long-term plan has been disclosed in the report which is as under:-

“23. That, it is submitted that in order to control industrial pollution from tannery sector in Jajmau area, State Government has devised a short term and a long term plan so that pollution of river Ganga due to effluent from tanneries is completely stopped.

*A. That as a **Short Term Plan**, the strategy adopted is the improvement in compliance through enforcement measures as detailed below:-*

a. That U.P. Pollution Control Board issued guidelines vide letter dated 08.06.2019 to assess the infrastructure of tannery units according to their permitted capacity in order to restrict tannery units to reduce the chances of overflow from pumping stations as well as excess effluent going into CETP.

b. That it is submitted that UPPCB has sealed 169 drums & 47 Paddles in 92 tannery units associated with Pumping Station-03 (Wajidpur), 54 drums & 12 Paddles in 19 tannery units associated with Pumping Station-04 (Budhiyaghat), 13 drums & 07 Paddles in 12 tannery units associated with Pumping Station-01 (Chhabilepurwa) to control excess flow generation.

- c. That U.P. Pollution Control Board restricted the overall processing capacity of each tannery unit further to 50% of its consented capacity in order to reduce the chances of excess flow going to pumping stations and subsequently to CETP inlet.
- d. That U.P. Pollution Control Board implemented roster of operation of tannery units attached to different pumping stations. As per this roster, tannery units attached to pumping station 1 and 3 are operated for first 15 days of the month and tannery units attached to pumping station 2 and 4 are operated for next 15/16 days of the month.
- e. That U.P. Pollution Control Board has informed that Surprise checks are conducted by UPPCB and Inter Departmental Teams as well as Third Party Institutions like IIT, HBTU, PCCRI, NEERI from time to time and actions are taken against the defaulter units as well as operator of CETP.
- f. That U.P. Pollution Control Board has imposed the environmental compensation against the defaulting units from time to time. Environmental Compensation of Rs 2.98 Crore imposed against 283 defaulter tannery units, out of which Environmental Compensation of Rs. 1.22 Crore has been deposited by 227 tannery units of Jajmau Kanpur.
- g. That U.P. Pollution Control Board has imposed Environmental Compensation against the operator of CETP and STPs at Jajmau Kanpur i.e. Ganga Pollution Control Unit, U.P. Jal Nigam, Kanpur / M/s Kanpur River Management Pvt Ltd, Kanpur from time to time on the basis of different defaults. The detail of imposition of Environmental Compensation imposed in year 2023 is enclosed herewith and marked as **Annexure No. -14.**

B. That as a **Long Term Plan**, the strategy adopted involves construction of a new CETP and upgradation of existing PETPs along with adoption of cleaner technology.

That it is submitted that in order to improve the treatment capacity of existing tannery units of Jajmau, Kanpur, Central Government has taken effective initiation and for Jajmau

Tannery cluster, National Mission for Clean Ganga, New Delhi has approved 20 MLD CETP (dilution based) worth Rs. 554 Crore, as per Hon’ble NGT direction. The proposal was consented by the tannery association and the Government of Uttar Pradesh. The member tannery units have committed to contribute Rs. 25 crore and has proposed to bear the estimated cost of Rs. 137 Crore for upgradation of their Primary Effluent Treatment Plants (PETPs). Rest 392 Crore will be borne by National Mission for Clean Ganga (NMCG). The Special Purpose Vehicle "Jajmau Tannery Effluent Treatment Association (JTETA)" of member tannery units and district administration headed by District Magistrate, Kanpur Nagar has been formed.

That the Jajmau Tannery Effluent Treatment Association (JTETA) has allotted the work of construction of CETP to M/s VA Tech Wa Bag Ltd. after finalizing the tender with approval from NMCG. NMCG has also appointed construction engineer to supervise the ongoing work and estimated time of completion was 13.08.2023 initially, which is now reported to be upto April 2024. The new CETP is being constructed in the same premises of existing CETP and the treated effluent is proposed to be used for irrigation after dilution with treated STP effluent. NMCG has also reviewed the progress of installation of 20 MLD CETP time to time with concerning stakeholders. As per the information provided by JTETA, vide letter dated 20.11.2023, the physical progress of the project as on 30.10.2023 is as below :

Components	%Civil	%Mechanical	%Electrical	%Instrumentation	%Cumulative
CETP	97%	74%	89%	60%	80%
CCRU	100%	100%	100%	100%	100%
ZLD	100%	90%	87%	57%	84%
C&C	89%	10%	15%	5%	30%
Total	97%	69%	73%	56%	73%

*The copy of letter dated 20.11.2023 sent by JTETA is enclosed herewith and marked as **Annexure No.-15.***

That after the construction of new 20 MLD CETP & allied structures like closed conveyance system to carry primarily treated effluent from member tannery units to CETP & CCRP and upgradation of existing PETP by tanneries, complete elimination of accidental discharge in river Ganga would be ensured.

*That U.P. Pollution Control Board has directed JTETA to submit monitorable ferti-irrigation plan vide letter dated 18.11.2023. Secretary, JTETA vide its letter dated 24.11.2023 has informed that the existing CETP and STP are together discharging the treated water of 214 MLD (Existing flow capacity 130 MLD+43 MLD+5 MLD+36 MLD) in the irrigation channel. As per the site survey, the existing irrigation channel has the capacity to withstand more than 240 MLD. Hence, the proposed CETP capacity of 20 MLD ensures that the treated water discharge will not exceed the capacity of the existing irrigation channel. This will help to maintain a sustainable and efficient irrigation system without causing any potential issues or overloading. Also, as per the contract issued by the NMCG, it has been clearly declared that the treated water has to be discharged into the existing irrigation channel. JTETA has informed that final Irrigation management plan compiled by a reputed/approved organization within next 45 days. The copy of letter dated 24.11.2023 send by JTETA is enclosed herewith and marked as **Annexure No.-16.**"*

12. The over-sight committee has also submitted the report in pursuance to the earlier orders passed in OA No. 985/2019 and OA No.986/2019 which discloses the status of M/s. Uttar Pradesh Waste Management Project (UPWMP), Kumbhi, Akbarpur, Kanpur Dehat and Bharat Oil and Waste Management Ltd.(BOWML), Kumbhi, Akbarpur, Kanpur Dehat. The inspection note of chromium dump site at Khanchandpur, Rania, Kanpur Dehat reveals that at the time of inspection yellowish soil was seen over the land surface of the site which indicated presence of chromium. The soil sample analysis report of that area shows concentration of hexavalent chromium beyond the permissible limits. The report of the three samples taken from the

different site by the oversight committee, as disclosed in paragraph no. 6, also shows presence of hexavalent chromium concentration beyond the permissible limit. Similar was the position noticed in respect of chromium dump site at Waris Chemical Pvt.ltd. Kanpur Dehat, Heliger Chemical Pvt. Ltd.

13. In the report, allegation of the representative of M/s. Unichem India has been noted that four units located in district Fatehpur at the border of Kanpur, hardly 30 kms away from the dump site at Rania namely, Ashiana Chemical Pvt. Ltd., Ashma Chemical Pvt. Ltd., Associated Chemicals and Madhuchandra Chemical Pvt.ltd. were never touched though they also produced the same hazardous waste and dumped it at the site. Hence, this issue needs to be duly considered by the concerned authority and report in respect of these four units is required to be submitted. The report contains point wise response to the issue raised in the letter of Ms. Pratiba Shukla. The response also reveals certain violations and consequential inaction by UPPCB.

14. During the course of argument, the Learned Counsel for the UPPCB has stated that there are no tanneries in Rania dump site and there are only seven manufacturing units, manufacturing chromium sulphate. So far as, Jajmau site is concerned there are 400 tanneries.

15. Learned Counsel for the applicant has submitted that the closed units are now being converted by the Project Proponents into school, charitable institutions, hospitals etc without restoration of soil, therefore, such activities on contaminated soil will have ill effect on the health of children, patients or persons using the premises, hence, it should be immediately stopped.

16. The Learned Counsel appearing for UPPCB has submitted that it lies within the jurisdiction of the state administrative authorities to stop such activities in the contaminated area. He has further submitted that the presence of Uttar Pradesh Ground Water Department is necessary to ensure the restoration of quality of ground water. He has submitted that all industries are now closed and there were only two clusters i.e. Jajmau Cluster and Unnao Banther Cluster where the tanneries were operating and there are about 10-15 individual tanneries operating outside the cluster.

17. On perusal of the previous orders, we find that though earlier the State of UP and UP Jal Nigam were represented before the Tribunal (order dated 08.02.2021 in OA 985/2019) but no one is present today on their behalf. Hence, at this stage, we direct the Counsel for the applicant to serve notice upon the State of UP, UP Jal Nigam and Irrigation Department and UP Ground Water Board of the State through their administrative heads for appearance and filing the response on or before the next date of hearing.

18. In OA No.528/2023- no report has been received. An issue has been raised that there is no piped water supply in the area concerned, therefore, water is supplied through the tankers which is also causing health hazard. Therefore, competent authority is directed to submit the action plan for supplying piped water in the area cornered.

19. The respondents are direct to submit time-bound plan for remediation.

20. Let the fresh report in terms of the above directions be submitted by the joint committee, UPPCB and State of UP within a period of 8 weeks by e-mail at judicial-ngt@gov.in preferably in the form of searchable PDF/ OCR Support PDF and not in the form of Image PDF.

21. A copy of this order be forwarded to the joint committee, UPPCB and State of UP by e-mail for compliance.

22. List the matter on 06.02.2024.

Prakash Shrivastava, CP

Sudhir Agarwal, JM

Arun Kumar Tyagi, JM

Dr. A. Senthil Vel, EM

November 29, 2023
Original Application No.985/2019,
Original Application No.986/2019 &
Original Application No.528/2023
JG